ASSIGNMENT 1:

1. Write a C++ program that takes in two integer values from the user and calculates the sum, difference, product, and quotient using the appropriate operators.
2. Write a C++ program that calculates the area of a circle given the radius as input from the user. Use the formula pi*r^2 and use the exponentiation operator to calculate the squared radius.
3. Write a C++ program that takes in three integer values from the user and checks if they are equal using the equality operator. If they are equal, print "Equal" to the screen, otherwise print "Not equal".
4. Write a C++ program that reads in a string from the user and checks if it is equal to the string "hello" using the equality operator. If it is equal, print "Hello!" to the screen, otherwise print "Goodbye!".
5. Write a C++ program that reads in two integer values from the user and checks if the first value is greater than the second using the greater than operator. If it is, print "First value is greater" to the screen, otherwise print "Second value is greater".
6. Write a C++ program that reads in a character from the user and checks if it is an uppercase letter using the logical AND operator and the isupper() function. If it is, print "Uppercase letter" to the screen, otherwise print "Not an uppercase letter".
7. Write a C++ program that reads in a string from the user and checks if it contains the character 'a' using the in operator. If it does, print "Contains 'a'" to the screen, otherwise print "Does not contain 'a'".
8. Write a C++ program that reads in an integer value from the user and checks if it is odd using the modulus operator. If it is odd, print "Odd number" to the screen, otherwise print "Even number".
9. Explain the difference between the assignment operator and the equality operator in C++. Give an example of each.
10. Write a C++ program that uses the ternary operator to determine the larger of two integers.
11. Explain the difference between the increment and decrement operators in C++. Give an example of each.
12. Write a C++ program that uses the logical AND operator to check if a number is both odd and divisible by 3.
13. Write a C++ program that uses the bitwise XOR operator to swap the values of two variables.
14. Explain the difference between the left shift and right shift operators in C++. Give an example of each.
15. Write a C++ program that uses the conditional operator to determine the larger of two floating point numbers.

16. Explain the difference between the bitwise NOT and logical NOT operators in C++. Give an example of each.

17. Write a C++ program that uses the bitwise AND operator to check if a number is a power of 2.

18. Write a C++ programming to convert binary to decimal.

19. Write a C++ programming calculator to convertbinary to decimal, binary to hexadecimal, binary to octadecimal.